Is Trackflation Real? A Time-Series Analysis of World Records

By Tyler Duke
October 25, 2026

Introduction

The term *trackflation* has emerged in running discussions to describe the seemingly rapid improvement in athletic performances, particularly in recent years. Advances in shoe technology, nutrition awareness, and access to coaching resources have all been suggested as possible contributors. But what can historical world-record data tell us about changes in athletic performance?

To explore this question, I collected world-record progression data from World Athletics, covering track-and-field events dating back to the year 1900. Rather than focusing on individual athletes or specific events, I examine the annual number of world records broken over time. Using time-series visualization, decomposition, and ARIMA forecasting, I explore historical patterns and investigate whether record-breaking activity exhibits a recurring four-year cycle associated with the Summer Olympic Games. This project applies time-series methods covered in my coursework at NC State's Institute of Advanced Analytics to a real-world question in one of my hobby passions.

Methods

The dataset contains individual world-record progression entries, including the event, athlete, performance, and date the record was established. I cleaned the data and aggregated the observations into annual counts of world records broken, creating a time series to explore historical trends and forecast future activity.

I then reorganized the annual observations into a quarterly time series, with each quarter representing one position in a four-year cycle. This allowed me to apply Seasonal and Trend decomposition using Loess (STL), a method covered in class that separates a time series into its trend, seasonal, and remainder components. By examining these components, I could explore whether the data contained a recurring pattern that might align with the Olympic cycle.

Finally, I fit an ARIMA model to the transformed series and generated forecasts for the next ten annual observations, extending the analysis through the next three Summer Olympic Games in 2028, 2032, and 2036. ARIMA modeling provides a way to use patterns in historical observations to estimate future values.

Results

The historical visualization reveals substantial variation in the number of world records broken from year to year [See Figure 1]. Marking the Olympic years provides a visual reference for investigating whether record-breaking activity follows a recurring four-year pattern. This type of visualization is useful for identifying patterns that may not be immediately apparent from individual records or summary statistics alone.

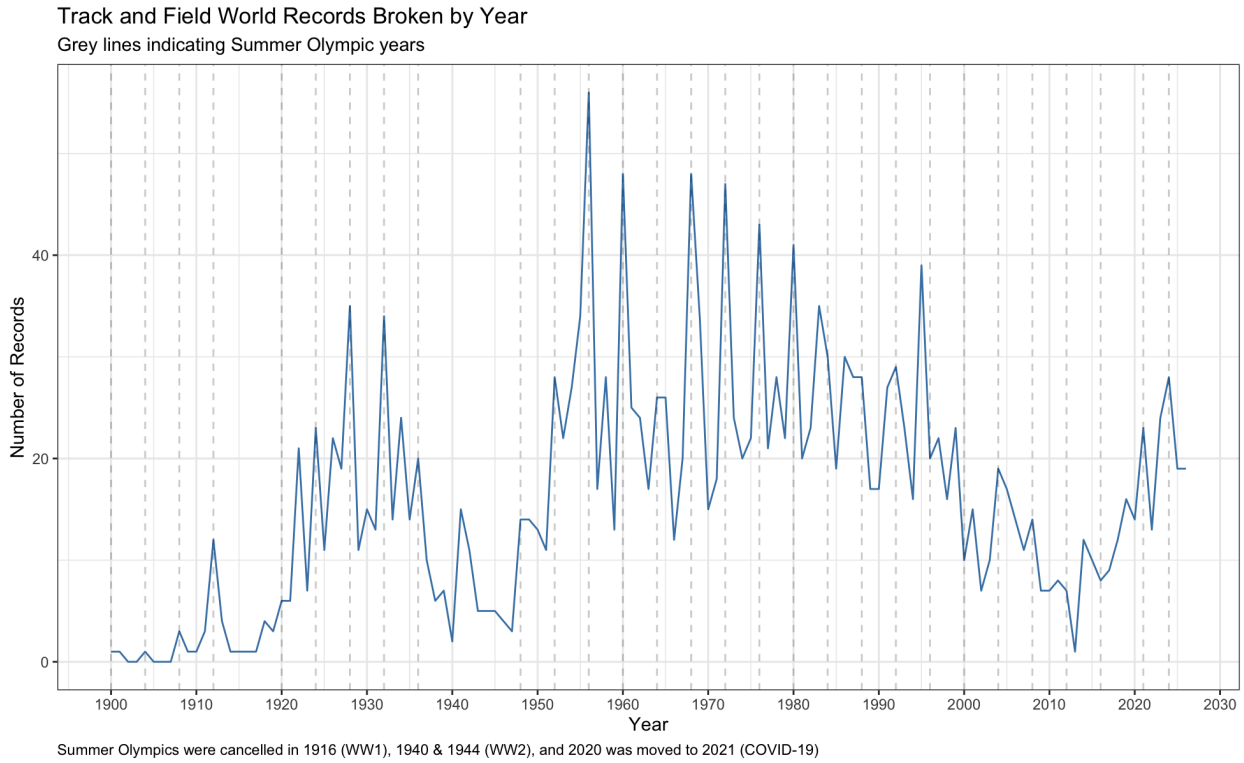


Figure 1: World Records by Year

The automatic model selection procedure identified an ARIMA(0,1,1)(1,0,0)[4] model as the best-fitting candidate. This model incorporates differencing to account for changes in the series level, a nonseasonal moving-average component, and an autoregressive component with a seasonal period of four observations. Because each position in the transformed series represents one calendar year, the seasonal period corresponds to a four-year cycle.

Applying this model allowed me to extend the analysis beyond describing historical data and investigate what future record-breaking activity might look like if the patterns captured by the model continue. I generated forecasts for the next ten annual observations, covering the period through the 2036 Summer Olympics [See Figure 2]. It is interesting to note that the Olympic seasonal pattern persists into the forecast period, indicating that while record-breaking activity varies over time, the underlying cyclical momentum remains consistent and does not point to a dramatic "trackflation" surge.

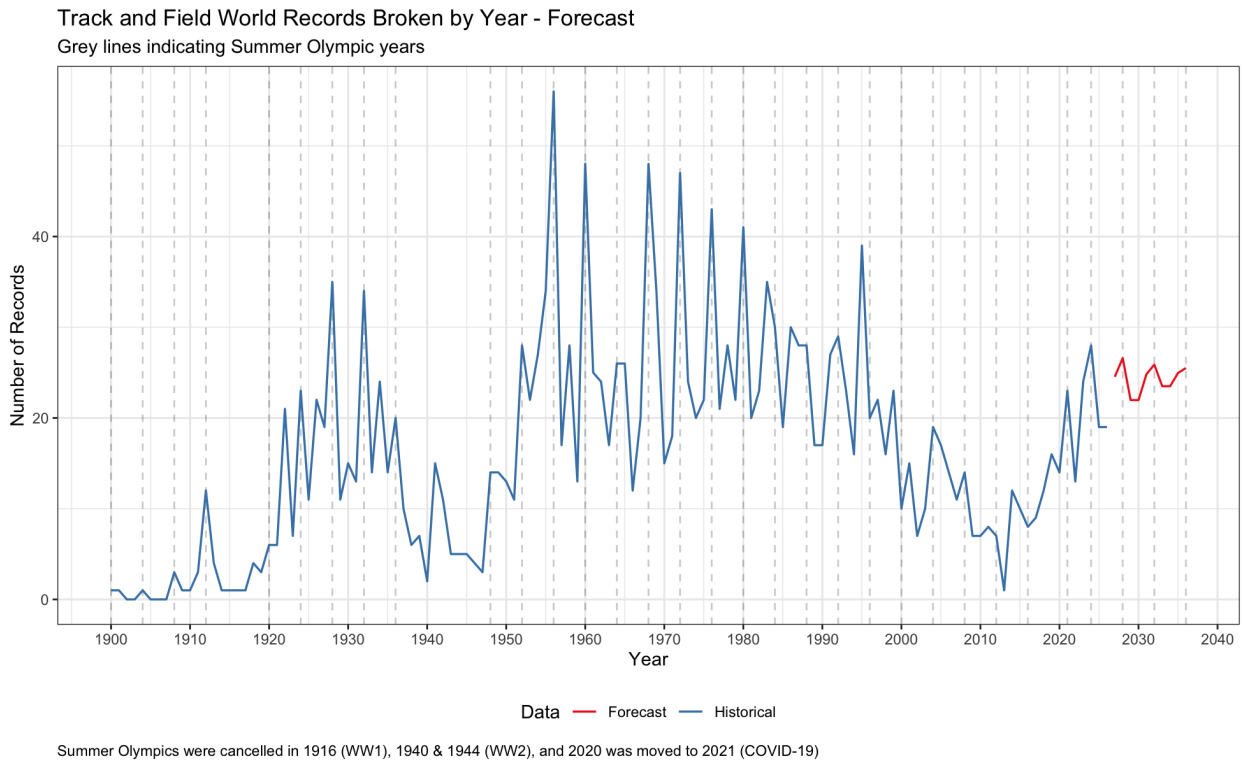


Figure 2: World Records by Year with Forecast

From a practical perspective, this approach illustrates how time-series modeling can be used to establish a historical baseline and evaluate whether future observations follow expected patterns. For example, sports analysts could use similar methods to monitor record-breaking activity over time and identify periods that warrant further investigation into changes in equipment, training, or competition.

Conclusion

This analysis demonstrates how time-series visualization, STL decomposition, and ARIMA forecasting can be applied to a real-world question in athletics. Rather than relying solely on recent performances, historical data provide a way to investigate whether record-breaking activity exhibits longer-term trends or recurring patterns.

Ultimately, this project demonstrates how statistical methods can help turn a broad question about track and field into a measurable analytical problem. While the forecasts provide a starting point for understanding future record-breaking activity, further analysis of individual performance improvements and technological changes would be needed to determine whether trackflation is real and to identify the factors that might explain those improvements.